

Product Summary

$V_{(BR)DSS}$	$R_{DS(ON)} \text{ max}$	I_D $T_A = +25^\circ\text{C}$
60V	69mΩ @ $V_{GS} = 10\text{V}$	4.3A
	100mΩ @ $V_{GS} = 4.5\text{V}$	3.5A

Description

This new generation MOSFET has been designed to minimize the on-state resistance ($R_{DS(on)}$) and yet maintain superior switching performance, making it ideal for high efficiency power management applications.

Applications

- Motor control
- Transformer driving switch
- DC-DC Converters
- Power management functions
- Uninterrupted power supply

Features

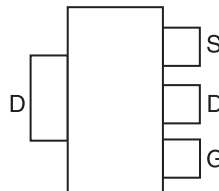
- 100% Unclamped Inductive Switch (UIS) test in production
- Fast switching speed
- Low on-resistance
- **Lead-Free Finish; RoHS compliant (Notes 1 & 2)**
- **Halogen and Antimony Free. "Green" Device (Note 3)**
- **Qualified to AEC-Q101 Standards for High Reliability**

Mechanical Data

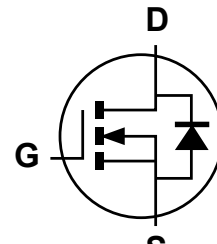
- Case: SOT223
- Case Material: Molded Plastic, "Green" Molding Compound. UL Flammability Classification Rating 94V-0 (Note 1)
- Moisture Sensitivity: Level 1 per J-STD-020
- Terminals Connections: See diagram below
- Terminals: Finish - Matte Tin annealed over Copper lead frame. Solderable per MIL-STD-202, Method 208 
- Weight: 0.112 grams (approximate)



Top View



Pin Out - Top View



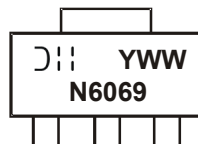
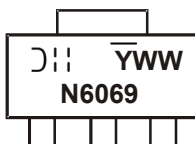
Equivalent Circuit

Ordering Information (Note 4)

Part Number	Qualification	Case	Packaging
DMN6069SE-13	Standard	SOT223	2,500 / Tape & Reel

- Notes:
1. EU Directive 2002/95/EC (RoHS) & 2011/65/EU (RoHS 2) compliant. All applicable RoHS exemptions applied.
 2. See http://www.diodes.com/quality/lead_free.html for more information about Diodes Incorporated's definitions of Halogen- and Antimony-free, "Green" and Lead-free.
 3. Halogen- and Antimony-free "Green" products are defined as those which contain <900ppm bromine, <900ppm chlorine (<1500ppm total Br + Cl) and <1000ppm antimony compounds.
 4. For packaging details, go to our website at <http://www.diodes.com/products/packages.html>.

Marking Information



 = Manufacturer's Marking
 N6069 = Marking Code
 YWW = Date Code Marking for SAT (Shanghai Assembly/ Test site)
 YWW = Date Code Marking for CAT (Chengdu Assembly/ Test site)
 Y or Y = Year (ex: 3 = 2013)
 WW = Week (01 - 53)

Maximum Ratings (@T_A = +25°C, unless otherwise specified.)

Characteristic		Symbol	Value	Units
Drain-Source Voltage		V _{DSS}	60	V
Gate-Source Voltage		V _{GSS}	±20	V
Continuous Drain Current (Note 6) V _{GS} = 10V	T _A = +25°C	I _D	4.3	A
	T _A = +70°C		3.3	
	T _C = +25°C	I _D	10	A
	T _C = +70°C		8	
Pulsed Drain Current (10μs pulse, duty cycle = 1%)		I _{DM}	25	A
Maximum Body Diode Continuous Current		I _S	3.2	A

Thermal Characteristics (@T_A = +25°C, unless otherwise specified.)

Characteristic		Symbol	Value	Units
Total Power Dissipation (Note 5)	T _A = +25°C	P _D	2.2	W
	T _A = +70°C		1.4	
Thermal Resistance, Junction to Ambient (Note 5)		R _{θJA}	58	°C/W
Total Power Dissipation (Note 5)		P _D	11	W
Thermal Resistance, Junction to Case (Note 5)		R _{θJC}	8.9	°C/W
Operating and Storage Temperature Range		T _J , T _{STG}	-55 to +150	°C

Electrical Characteristics (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Min	Typ	Max	Unit	Test Condition
OFF CHARACTERISTICS (Note 7)						
Drain-Source Breakdown Voltage	BV _{DSS}	60	—	—	V	V _{GS} = 0V, I _D = 250µA
Zero Gate Voltage Drain Current	I _{DSS}	—	—	1	µA	V _{DS} = 60V, V _{GS} = 0V
Gate-Source Leakage	I _{GSS}	—	—	±100	nA	V _{GS} = ±20V, V _{DS} = 0V
ON CHARACTERISTICS (Note 7)						
Gate Threshold Voltage	V _{GS(th)}	1	—	3	V	V _{DS} = V _{GS} , I _D = 250µA
Static Drain-Source On-Resistance	R _{DS(on)}	—	47	69	mΩ	V _{GS} = 10V, I _D = 3A
		—	54	100		V _{GS} = 4.5V, I _D = 2.4A
Diode Forward Voltage	V _{SD}	—	0.8	1.1	V	V _{GS} = 0V, I _S = 2.5A
DYNAMIC CHARACTERISTICS (Note 8)						
Input Capacitance	C _{iss}	—	825	—	pF	V _{DS} = 30V, V _{GS} = 0V f = 1MHz
Output Capacitance	C _{oss}	—	40	—		
Reverse Transfer Capacitance	C _{rss}	—	29	—		
Gate Resistance	R _G	—	2.3	—	Ω	V _{DS} = 0V, V _{GS} = 0V, f = 1.0MHz
Total Gate Charge (V _{GS} = 4.5V)	Q _g	—	7.2	—	nC	V _{DS} = 30V, I _D = 12A
Total Gate Charge (V _{GS} = 10V)	Q _g	—	16	—		
Gate-Source Charge	Q _{gs}	—	3.2	—		
Gate-Drain Charge	Q _{gd}	—	2.8	—		
Turn-On Delay Time	t _{D(on)}	—	3.8	—	nS	V _{DD} = 30V, V _{GS} = 10V, R _G = 6Ω, I _D = 12A
Turn-On Rise Time	t _r	—	6.7	—		
Turn-Off Delay Time	t _{D(off)}	—	16	—		
Turn-Off Fall Time	t _f	—	5.3	—		

- Notes:
- Device mounted on FR-4 substrate PC board, 2oz copper, with minimum recommended pad layout.
 - Device mounted on FR-4 substrate PC board, 2oz copper, with thermal vias to bottom layer 1inch square copper plate
 - Short duration pulse test used to minimize self-heating effect.
 - Guaranteed by design. Not subject to product testing.

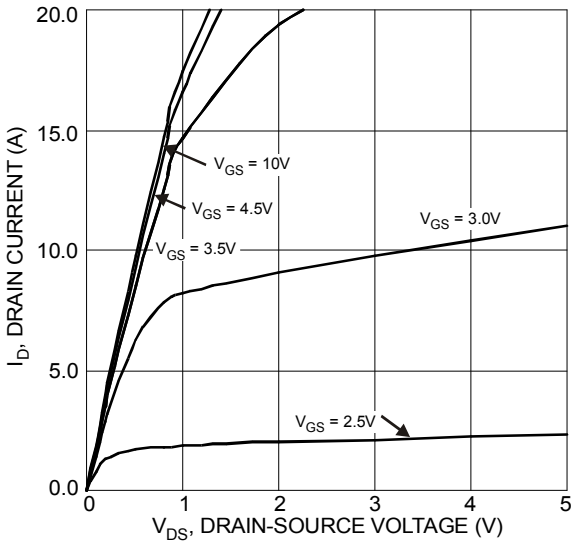


Figure 1 Typical Output Characteristic

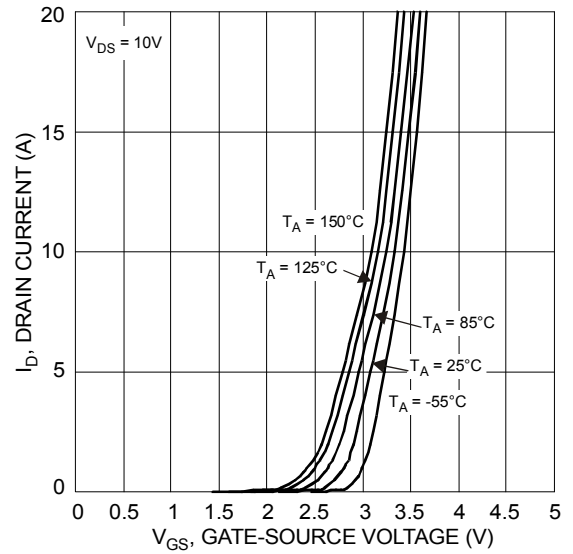


Figure 2 Typical Transfer Characteristics

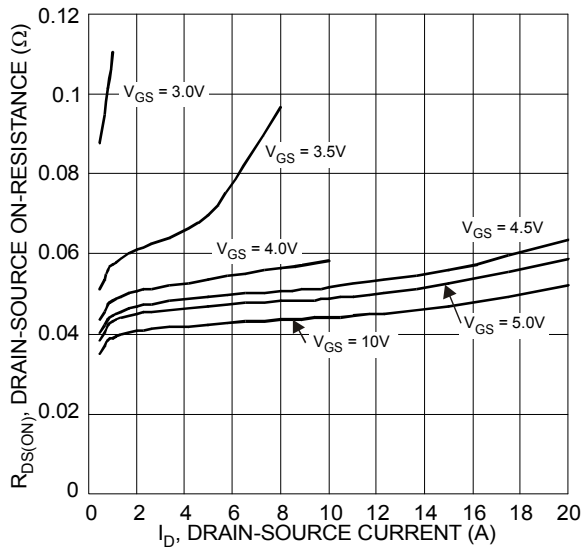


Figure 3 Typical On-Resistance vs. Drain Current and Gate Voltage

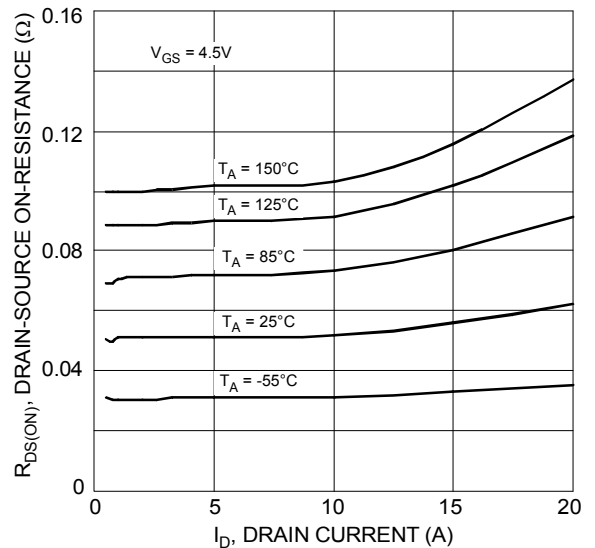


Figure 4 Typical On-Resistance vs. Drain Current and Temperature

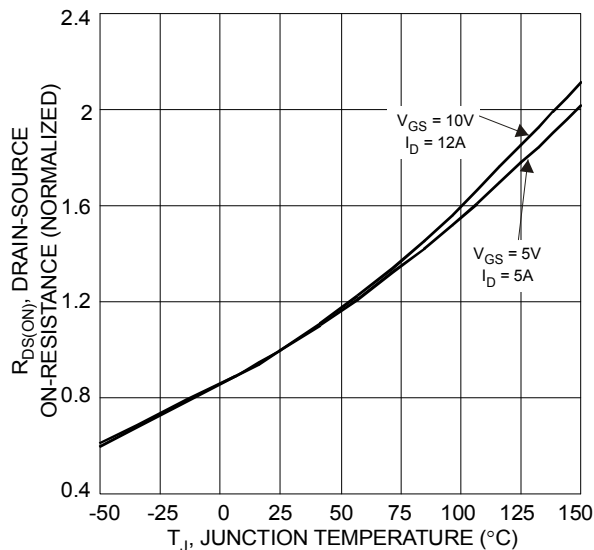


Figure 5 On-Resistance Variation with Temperature

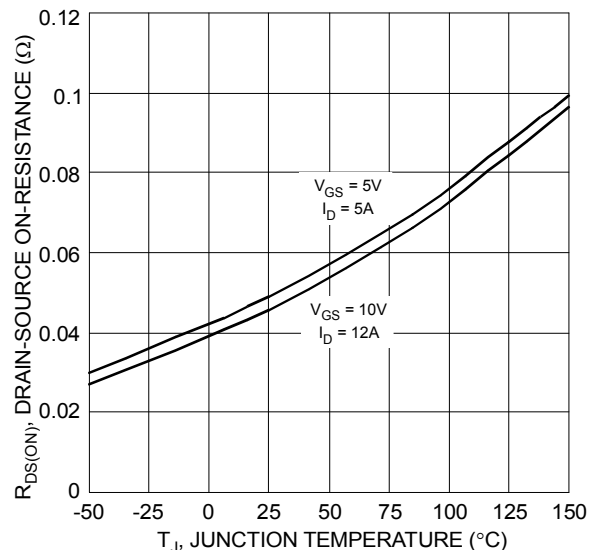


Figure 6 On-Resistance Variation with Temperature

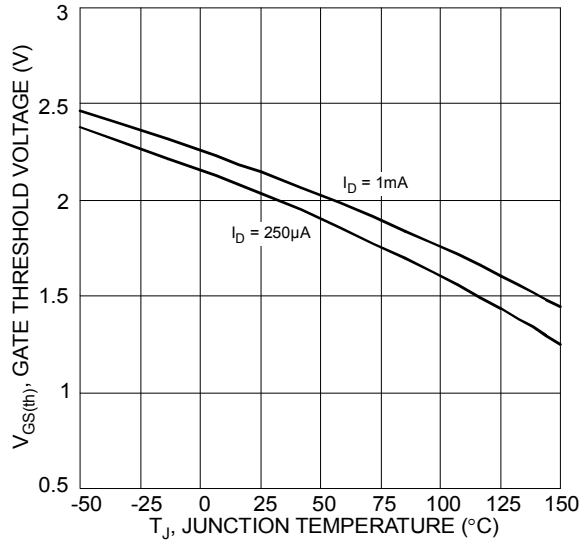


Figure 7 Gate Threshold Variation vs. Ambient Temperature

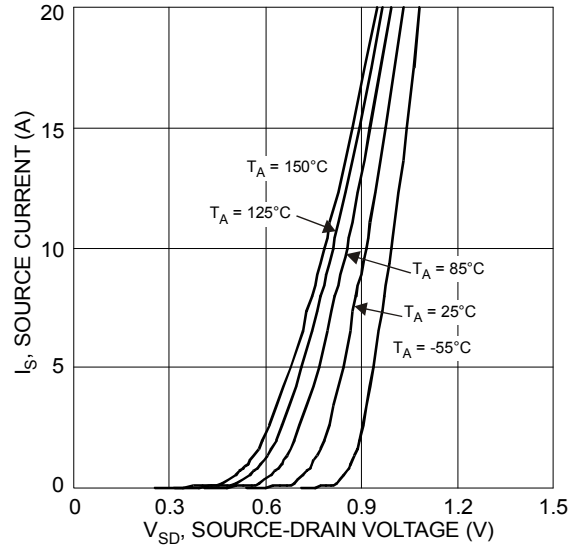


Figure 8 Diode Forward Voltage vs. Current

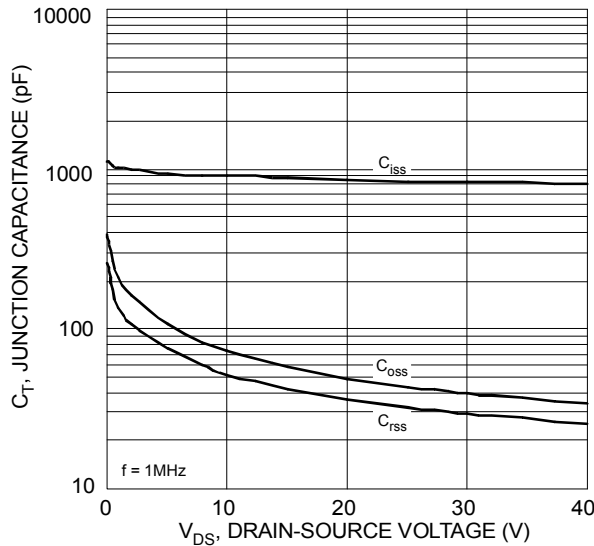


Figure 9 Typical Junction Capacitance

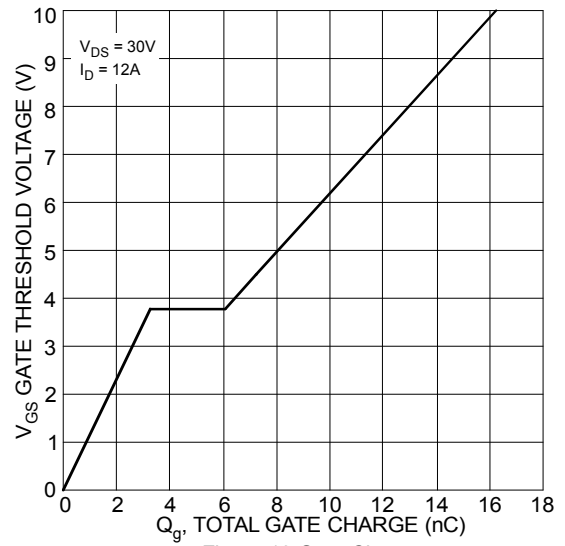


Figure 10 Gate Charge

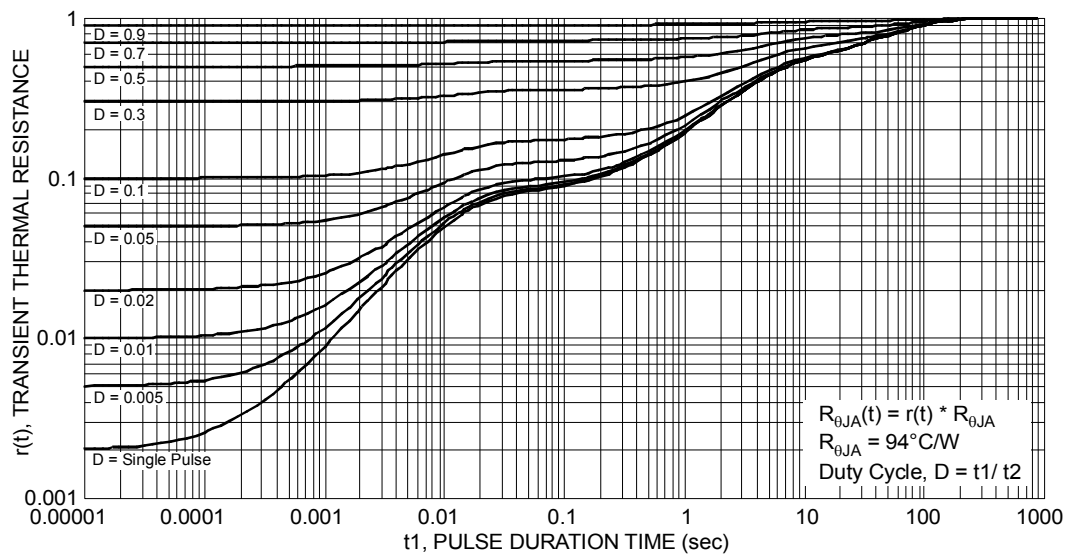
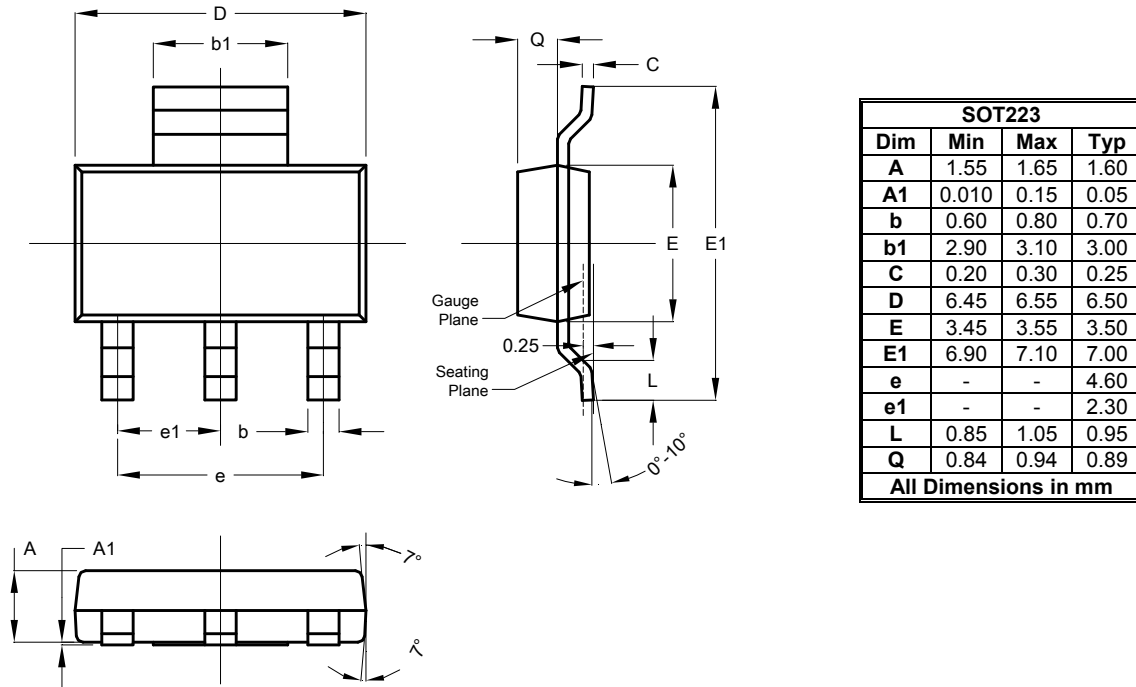


Figure 11 Transient Thermal Resistance

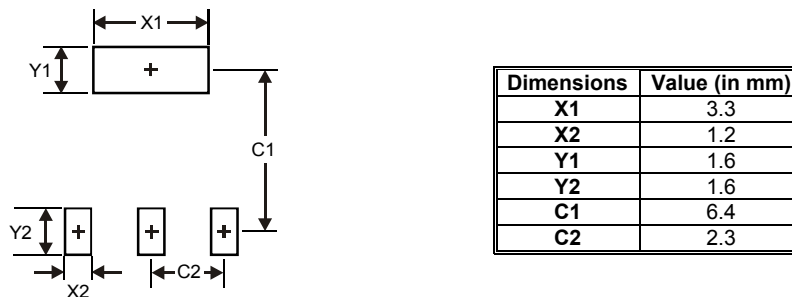
Package Outline Dimensions

Please see AP02002 at <http://www.diodes.com/datasheets/ap02002.pdf> for latest version.



Suggested Pad Layout

Please see AP02001 at <http://www.diodes.com/datasheets/ap02001.pdf> for the latest version.



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